HWA CHONG INSTITUTION
Preliminary Examination
Higher 1

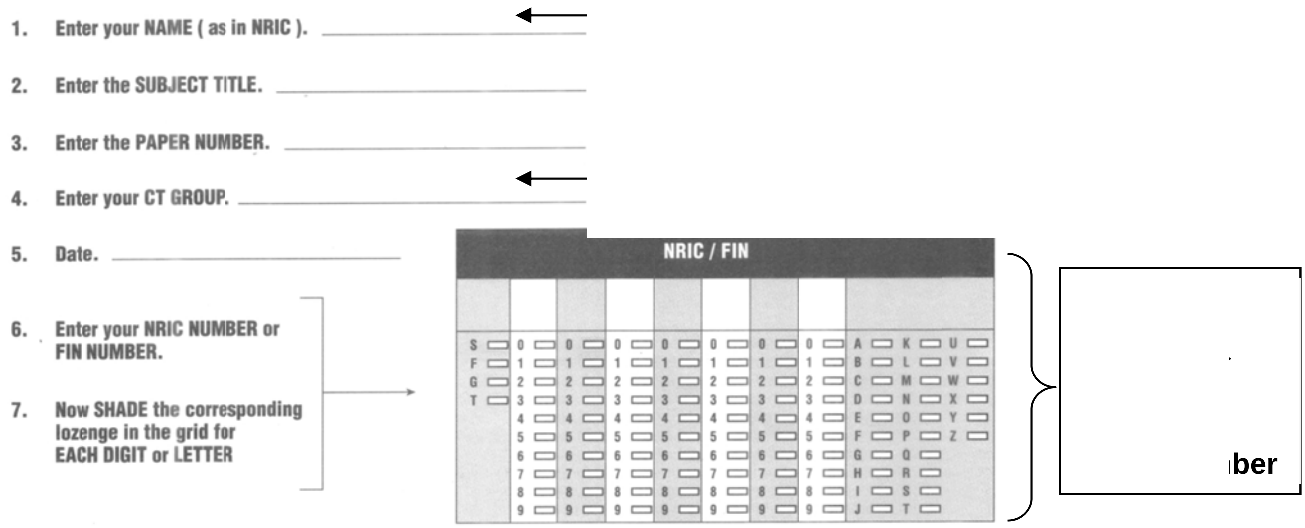
CANDIDATE NAME		**CT GROUP**	**12S**
CENTRE NUMBER		**INDEX NUMBER**	

CHEMISTRY **8872/01**
Paper 1 Multiple Choice **25 September 2013**
50 min

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Complete the information on the optical mark sheet (OMS) as shown below.



There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

Each correct answer will score one mark. A mark will **not** be deducted for a wrong answer.
Any rough working should be done in this booklet.

Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the one you consider to be correct.

1 How many neutrons are present in 0.13g of ^{13}C?
[L = the Avogadro constant]

A 0.06L **B** 0.07L **C** 0.13L **D** 0.91L

2 Phosphorus sulfide, P_4S_3, is used in small amounts in the tips of matches. On striking a match, this compound burns to produce an oxide of phosphorus in the +5 oxidation state and an oxide of sulfur in the +4 oxidation state.

How many moles of oxygen gas are needed to burn one mole of P_4S_3 in this way?

A 6 **B** 7.5 **C** 8 **D** 16

3 25.00 cm^3 of a solution of acidified iron(II) sulfate, $FeSO_4$, were titrated with 0.0200 mol dm^{-3} potassium manganate(VII). The mean titre was 27.40 cm^3.

The equation for the reaction is shown.

$$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$$

What is the concentration of iron(II) sulfate solution?

A 4.38×10^{-3} mol dm^{-3}

B 2.19×10^{-2} mol dm^{-3}

C 9.12×10^{-2} mol dm^{-3}

D 1.10×10^{-1} mol dm^{-3}

4 Which factor helps to explain why the first ionisation energies of the Group I elements decrease from lithium to sodium to potassium to rubidium?

A The nuclear charge of the elements increases.

B The outer electron is in an ‘s’ subshell.

C The repulsion between spin–paired electrons increases.

D The shielding effect of the inner shells increases.

5 In which pair of molecules are the values of the bond angles the closest?

A BF_3 and NH_3

B C_2H_4 and BF_3

C H_2O and C_2H_4

D CH_4 and H_2O

6 In the last century the Haber process was sometimes run at pressures of 1000 atm and higher. Now it is commonly run at pressures below 100 atm.

What is the reason for this change?

A An iron catalyst is used.

B Maintaining the higher pressures is more expensive.

C The equilibrium yield of ammonia is increased at lower pressures.

D The rate of the reaction is increased at lower pressures.

7 In the analysis of an oxide of nitrogen, 0.10 mol of the oxide were reacted with excess hydrogen under suitable conditions.

$$N_xO_y \xrightarrow{H_2(g)} xNH_3 + yH_2O$$

3.6 g of water were formed in this reaction, while the ammonia produced required 100 cm^3 of 1.0 mol dm^{-3} $HCl(aq)$ for neutralisation.

What is the formula of the oxide of nitrogen analysed?

A NO_2 **B** NO **C** N_2O **D** N_2O_5

8 In the gas phase, ammonia reacts with chlorine.

$$8NH_3 + 3Cl_2 \rightarrow N_2 + 6NH_4Cl$$

Which row indicates the correct combination of statements about this reaction?

	ammonia acts as a reducing agent	ammonia acts as a base	a dative bond is formed
A	✓	✓	✓
B	✓	×	×
C	×	✓	✓
D	×	×	✓

9 In which change would *only* van der Waals' forces have to be overcome?

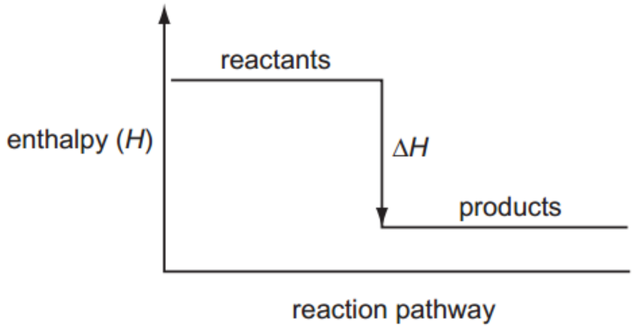
A evaporation of ethanol $C_2H_5OH(l) \rightarrow C_2H_5OH(g)$

B melting of ice $H_2O(s) \rightarrow H_2O(l)$

C melting of solid carbon dioxide $CO_2(s) \rightarrow CO_2(l)$

D solidification of butane $C_4H_{10}(l) \rightarrow C_4H_{10}(s)$

10 Which enthalpy change could **not** be correctly represented by the enthalpy diagram shown?



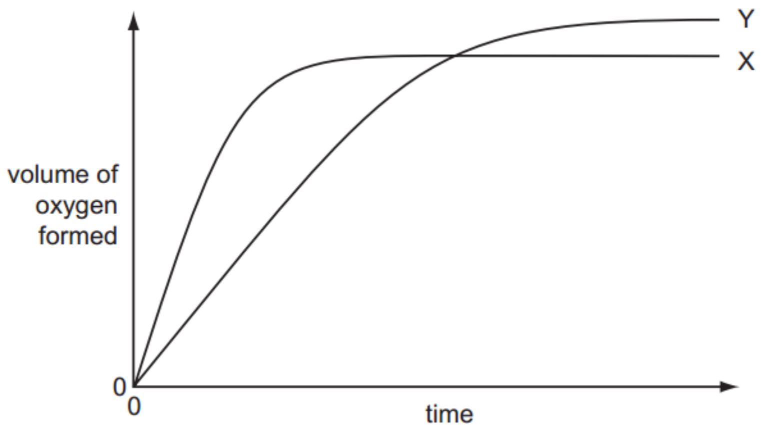
A standard enthalpy change of atomisation

B standard enthalpy change of combustion

C standard enthalpy change of hydration

D standard enthalpy change of neutralisation

11 In the diagram, curve **X** was obtained by observing the decomposition of 100 cm^3 of 1.0 mol dm^{-3} hydrogen peroxide, catalysed by manganese(IV) oxide.



Which alteration to the original experimental conditions would produce curve **Y**?

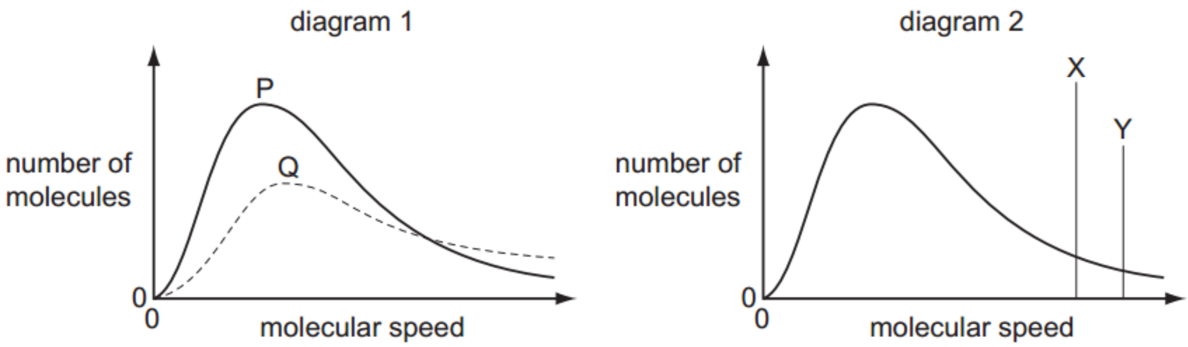
A adding some 0.1 mol dm^{-3} hydrogen peroxide

B adding water

C lowering the temperature

D using less manganese(IV) oxide

12 Different Boltzmann distributions are shown in the diagrams.



In diagram 1, one curve **P** or **Q** corresponds to a temperature higher than that of the other curve.

In diagram 2, one line **X** or **Y** corresponds to the activation energy for a catalysed reaction and the other line corresponds to the activation energy of the same reaction when uncatalysed.

Which combination gives the correct curve and line?

	higher temperature	presence of catalyst
A	**P**	**X**
B	**P**	**Y**
C	**Q**	**X**
D	**Q**	**Y**

13 Which set of solid elements contains a simple molecular structure, a giant covalent (macromolecular) structure and a giant metallic structure?

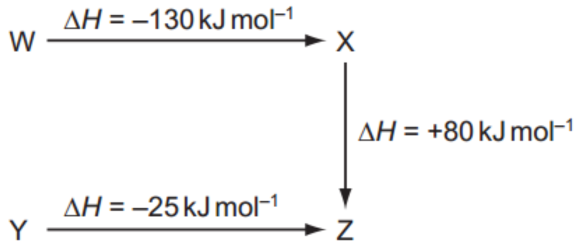
A Mg, P, S

B P, Si, C

C S, P, Si

D S, Si, A*l*

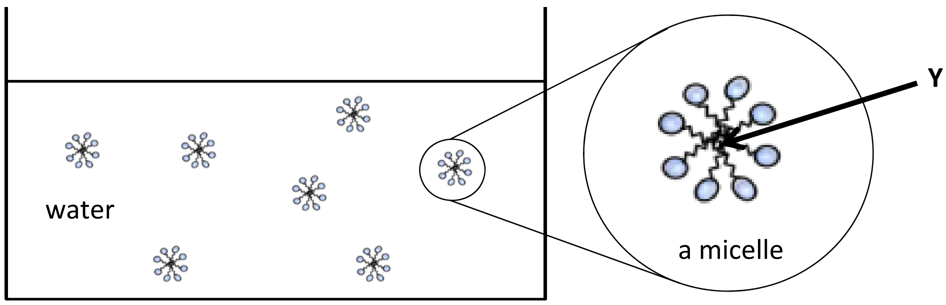
14 The diagram represents the energy changes for some reactions.



What are the natures of the conversions W → Y, Y → X and Z → W?

	W → Y	Y → X	Z → W
A	exothermic	endothermic	endothermic
B	exothermic	exothermic	endothermic
C	endothermic	exothermic	exothermic
D	endothermic	endothermic	exothermic

15 Casein is a protein commonly found in cow's milk. It exists in milk as a suspension of particles where molecules will group together in three–dimensional spherical clusters. Hydrocarbon chains fill the interior of the cluster and the ionic ends compose the outer surface. These molecular conglomerations are known as *micelles*.



What type of bonding is most likely to occur in the region labeled **Y**?

A Dispersion forces

B Hydrogen bonding

C Ion–dipole interactions

D Permanent dipole–permanent dipole interactions

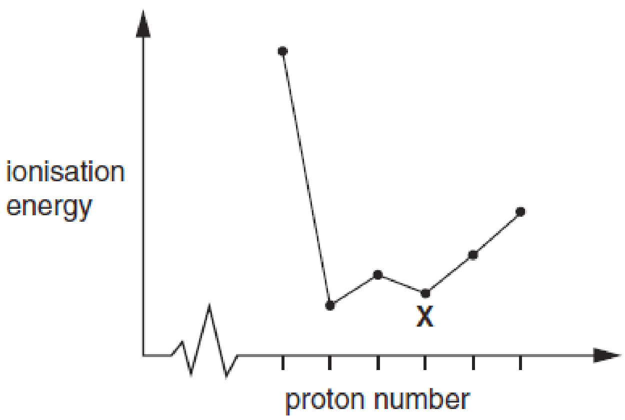
16 Three substances, **R**, **S** and **T**, have physical properties as shown.

substance	**R**	**S**	**T**
mp / °C	801	2852	3550
bp / °C	1413	3600	4827
electrical conductivity of solid	poor	poor	good

What could be the identities of **R**, **S** and **T**?

	R	**S**	**T**
A	MgO	NaC*l*	C [graphite]
B	MgO	NaC*l*	SiO_2
C	NaC*l*	MgO	C [graphite]
D	NaC*l*	MgO	SiO_2

17 The sketch below shows the variation of first ionisation energy with proton number for six elements of consecutive proton numbers between 1 and 18 (H to Ar).



What is the identity of the element **X**?

A Mg **B** A*l* **C** Si **D** P

18 When 0.20 mol of hydrogen gas and 0.15 mol of iodine gas are heated at 723 K until equilibrium is established, the equilibrium mixture is found to contain 0.26 mol of hydrogen iodide.

The equation for the reaction is as follows.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

What is the correct expression for the equilibrium constant K_c?

A $\dfrac{2 \times 0.26}{0.20 \times 0.15}$

B $\dfrac{(2 \times 0.26)^2}{0.20 \times 0.15}$

C $\dfrac{(0.26)^2}{0.07 \times 0.02}$

D $\dfrac{(0.26)^2}{0.13 \times 0.13}$

19 In which class of compound, in its general formula, is the ratio of hydrogen atoms to carbon atoms the highest?

A alcohols

B aldehydes

C carboxylic acids

D halogenoalkanes

20 The Russian composer Borodin was also a research chemist who discovered a reaction in which two ethanal molecules combine to form a compound commonly known as aldol (reaction I). Aldol forms another compound on heating (reaction II).

I $2CH_3CHO \rightarrow CH_3CH(OH)CH_2CHO$

II $CH_3CH(OH)CH_2CHO \rightarrow CH_3CH{=}CHCHO + H_2O$

Which of the following best describes reactions I and reaction II?

	I	I
A	addition	elimination
B	addition	reduction
C	elimination	reduction
D	substitution	elimination

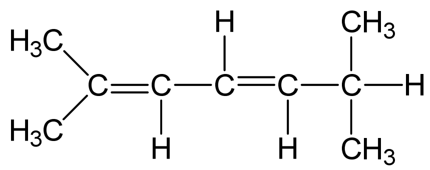
21 A hydrocarbon, **R** on heating with an excess of hot concentrated acidic $KMnO_4$(aq), produces CH_3COCH_3 and $(CH_3)_2CHCO_2H$ as the only organic products.

What is **R**?

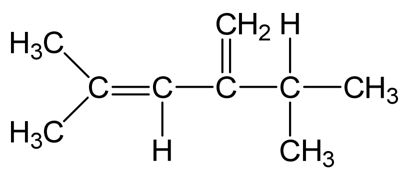
A $(CH_3)_2C{=}C(CH_3)CH_2CH_3$

B $(CH_3)_2C{=}CHCH_2CH_2CH_3$

C



D



22 Which property of benzene may be directly attributed to the stability associated with its delocalised electrons?

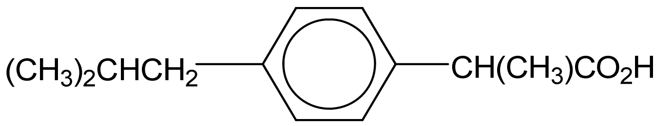
A It has a low boiling point.

B It does not conduct electricity.

C It tends to undergo substitution rather than addition reactions.

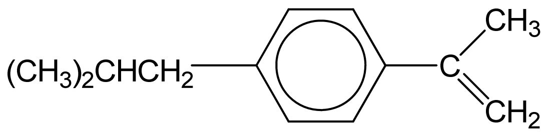
D It does not react with acidified $KMnO_4$

23 Ibuprofen is an anti–inflammatory drug.

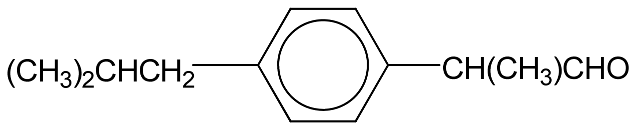


ibuprofen

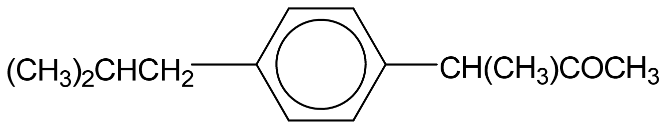
What reaction would lead to its formation?

A 

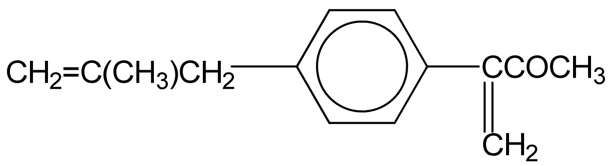
+ hot concentrated $KMnO_4$

B 

+ warm acidified $K_2Cr_2O_7$

C 

+ warm H_2SO_4(aq)

D 

+ H_2 / Pt catalyst

24 In its reaction with sodium, 1 mol of a compound **X** gives 1 mol of H_2(g).

Which compound might **X** be?

A $CH_3CH_2CH_2CH_2OH$

B $(CH_3)_3COH$

C $CH_3CH_2CH_2CO_2H$

D $CH_3CH(OH)CO_2H$

25 A compound **P** was boiled with aqueous sodium hydroxide and the resulting mixture was cooled and acidified with dilute sulfuric acid. The final products included a compound $C_3H_6O_2$ and an alcohol which gave a positive triiodomethane test.

Which of the following formulae could represent **P**?

A $CH_3CH_2CO_2CH_3$

B $CH_3CH_2OCOCH_3$

C $CH_2(OH)CH_2CH_2COCH_3$

D $CH_3CH_2CO_2CH_2CH_3$

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, 2 and 3 are correct	1 and 2 are correct	2 and 3 are correct	1 only is correct

No other combination of statements is used as a correct response.

26 *Use of the Data Booklet is relevant to this question.*

In which pairs do both species have the same number of unpaired p electrons?

1 O and Cl^+

2 F^+ and Ga^+

3 P and Ne^+

27 Which are features of the structure of metallic copper?

1 a lattice of ions

2 delocalised electrons

3 ionic bonds

28 Which descriptions of the ammonium ion are correct?

1 It contains ten electrons.

2 It has a bond angle of 109.5°.

3 It has four bonding pairs of electrons.

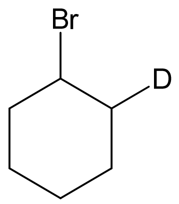
The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, 2 and 3 are correct	1 and 2 are correct	2 and 3 are correct	1 only is correct

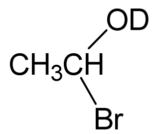
No other combination of statements is used as a correct response.

29 Deuterium, D, is the ${}^{2}_{1}H$ isotope of hydrogen. DBr has the same chemical properties as HBr.

Which compounds could be made by the reaction of DBr with another compound in a single reaction?

1 

2 $CHBr_2CHDBr$

3 

30 How can the rate of reaction between ethanal and aqueous hydrogen cyanide be increased?

1 by irradiation with ultraviolet light

2 by a rise in temperature

3 by the addition of a small quantity of aqueous sodium cyanide